Energy Resource Detective Chart

Grade 4 reference · every resource has benefits AND limitations

Name: _____ **Date:** _____

PAGE 1 — NONRENEWABLE RESOURCES

Coal *NONRENEWABLE · solid*
Renewable / Nonrenewable: Nonrenewable (solid)
How it is used: Burned to heat homes, cook, and make electricity
Energy pathway: burn coal → heat water → steam → turbine → generator → electricity
Benefit: Abundant and inexpensive
Limitation: Mining is dangerous; burning releases substances into the air
Works well when…: A lot of coal is available and inexpensive

Petroleum (crude oil) *NONRENEWABLE · liquid*
Renewable / Nonrenewable: Nonrenewable (liquid)
How it is used: Refined into gasoline, diesel, kerosene, plastics, and more
Energy pathway: burn fuel → heat → move engine parts (or make electricity)
Benefit: Powers vehicles; makes many products
Limitation: Drilling and spills can harm land, water, and wildlife
Works well when…: Liquid fuel is needed for engines and products

Natural gas *NONRENEWABLE · gas*
Renewable / Nonrenewable: Nonrenewable (gas)
How it is used: Burned for heating and cooking; piped into homes
Energy pathway: burn gas → heat
Benefit: Heats homes; cooks food; burns cleaner than coal
Limitation: Flammable; pipes can break and leak
Works well when…: Pipelines can reach homes and buildings

Uranium *NONRENEWABLE · mined*
Renewable / Nonrenewable: Nonrenewable (mined)
How it is used: Heats water in a nuclear power plant
Energy pathway: nuclear heat → steam → turbine → generator → electricity
Benefit: Reliable electricity with little air pollution
Limitation: Used fuel stays dangerous a long time and must be stored safely
Works well when…: Reliable electricity is needed

PAGE 2 — RENEWABLE RESOURCES

Wind *RENEWABLE*
Renewable / Nonrenewable: Renewable
How it is used: Turns turbine blades to make electricity
Energy pathway: moving air → blades → generator → electricity
Benefit: Renewable; no fuel burned
Limitation: Wind is not always blowing; needs a windy place
Works well when…: The wind blows steadily and strongly

Hydroelectric (moving water) *RENEWABLE*
Renewable / Nonrenewable: Renewable
How it is used: Water through a dam spins a turbine
Energy pathway: water in reservoir → dam → turbine → generator → electricity
Benefit: Renewable; water can be stored (predictable)
Limitation: Dam floods land; can block fish
Works well when…: There is a large, steady river

Solar *RENEWABLE*
Renewable / Nonrenewable: Renewable
How it is used: Sunlight makes electricity or heats water and food
Energy pathway: sunlight → solar cell → electricity
Benefit: Renewable; free sunlight; no fuel burned
Limitation: Depends on sunlight (location, weather, time of day)
Works well when…: A place gets steady, strong sunlight

Geothermal (Earth's heat) *RENEWABLE*
Renewable / Nonrenewable: Renewable
How it is used: Heat from underground warms buildings or makes steam for electricity
Energy pathway: Earth's heat → steam → turbine → generator → electricity
Benefit: Little pollution; heat is always underground
Limitation: Needs a special (volcanic) location; costly to build
Works well when…: There is hot rock near the surface

KEY IDEA **Remember: abundant ≠ renewable · renewable ≠ available everywhere · every resource has benefits AND limitations.**